

KIHYUN HAN

423 Thomas Building, University Park, PA 16802, USA

Email: khnhan@psu.edu [Github] [Website]

RESEARCH INTERESTS

Semiparametric modeling, Nonparametric modeling, Survival analysis, Robust inference, Predictive inference, Distribution shift.

EDUCATION

Ph.D. Statistics

August 2022 - May 2027 (*expected*)

Pennsylvania State University, University Park, PA

- Committee: Yanyuan Ma (Chair), Cory McCartan, Bing Li, Keisuke Hirano
- Dissertation: *Semiparametric and Nonparametric Inference under a Right-Censored Covariate*.
- Comprehensive exam: June 2025
- Qualifying exam: May 2023
[passed with distinction in all three exams (Theory 1, 2, and Applied)]

B.S. Mathematical Sciences, B.S. Statistics (double major)

March 2016 - August 2022

Seoul National University, Seoul, South Korea

- On leave for military service in Korean army for 2 years (2019 - 2020)
- Summa Cum Laude

RESEARCH

Preprints

* denotes equal contribution.

1. **Han, K.**, Ma, Y., Marder, K., & Garcia, T. (2026).
SPYCE: A Doubly Robust Estimator for Trials Targeting Early Huntington Disease under Outcome-Dependent Censoring. [arXiv]
2. **Han, K.**, Ma, Y., Marder, K., & Garcia, T. (2026).
PRESCO: Efficient Prediction Intervals under a Right-Censored Covariate. [arXiv]
3. Li, M.* , **Han, K.***, & Ma, Y. (2026).
From Matrix Inversion to Constraints: Provably Tighter Confidence Regions for Importance Weights in Label Shift. [arXiv]

Working Papers

* denotes equal contribution.

4. **Han, K.**, Ma, Y., Marder, K., & Garcia, T. (2026+).
Nonparametric Monotonicity-Preserving Quantile Regression under a Right-Censored Covariate.
5. **Han, K.***, Li, M.* , & Ma, Y., (2026+).
Honest Conformal Prediction under Label Shift with Quantile Optimization.
6. **Han, K.**, & McCartan, C. (2026+).
Prediction Sequence: Anytime-Valid Regression-Based Sequence of Prediction Intervals.

SOFTWARE

R packages

1. `spyce`: Estimating a regression parameter in right-censored covariate setting under outcome-dependent censoring.
2. `prescco`: Producing a prediction interval under a right-censored covariate.

PRESENTATIONS

Contributed Talks

- **Penn State SMAC Talk.**
Prediction Sequence: Anytime-Valid Regression-Based Sequence of Prediction Intervals.
Department of Statistics, Penn State University. University Park, PA September 2026
- **2026 Joint Statistical Meeting.**
Semiparametric Prediction under Right-Censored Covariates.
Improving and Assessing Risk Prediction Models Session. Boston, MA August 2026
- **2025 Joint Statistical Meeting.**
SPYCE: Semi-Parametric Y-dependent right-Censored Covariate Estimator.
Semiparametric Modeling Advanced Research Session. Nashville, TN August 2025

TEACHING

Instructor

- STAT414: Introduction to Probability Theory Summer 2025

Teaching Assistant

- STAT544: Categorical Data Analysis (*Ph.D.-level*) Fall 2024
- STAT200: Elementary Statistics (*lab instructor*) Fall 2023, Spring 2024
- STAT414: Introduction to Probability Theory Fall 2022, Spring 2023, Summer 2023

AWARDS/SCHOLARSHIPS/MEMBERSHIPS

- Member of Mu Sigma Rho: The National Statistics Honorary Society April 2025 - Present
- Harkness Statistics Enhancement Fund Award (best first-year Ph.D. student award),
Department of Statistics, Penn State University June 2023
- 3rd place in Korea University Students Industrial Engineering Project Competition,
Korean Institute of Industrial Engineers November 2021
- Gold Prize in University Students Contest of Mathematics (for math major),
Korean Mathematical Society November 2018
- Presidential Science Scholarship,
Korea Student Aid Foundation March 2016 - August 2022

SERVICE

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| Journal Reviewer | Stat | 2024 - Present |
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